Skew Brownian Motion and Related Diffusions

Dedicated to Albert N. Shiryaev on the occasion of his 90th Birthday

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Abstract

We review the construction, characterizations, descriptions, and properties of the scalar diffusion known as *Skew Brownian Motion.* We highlight the crucial rôle semimartingale local time plays in understanding the nature of this diffusion, and point out relations to other such random motions, including that of two particles with independent driving Brownian Motions and dispersions assigned by rank, as well as "spider" and "sticky" Brownian Motions.

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1 Introduction

In their inimitable, telegraphic style, Itô & McKean [20] introduce (Problem 1, straddling pages 115 -116) a scalar diffusion they call *Skew Brownian Motion:* one considers the excursions away from the origin of a reflected Brownian motion, and changes the sign of each excursion with probability $1-\alpha$, independently from excursion to excursion. Each given excursion remains positive with probability $\alpha \in (0,1)$, and becomes negative with probability $1-\alpha$. If $\alpha = 1/2$, a standard Brownian motion emerges; if $\alpha \neq 1/2$, one obtains a diffusion (continuous, strong Markov) process. This, and every continuous process $X(t)$, $0 \le t < \infty$ with the same finite-dimensional distributions, is called *Skew Brownian Motion* with skewness parameter α.

A detailed and rigorous implementation of the above construction is carried out by Prokaj [32]; see also [18]. The resulting process behaves just like Brownian motion when away from the origin; but passes the origin more easily in one direction than it does in the other, when $\alpha \neq 1/2$. This asymmetry is quantified via the local time relation (21) below. Think of a particle diffusing through a semi-permeable membrane under osmotic pressure; cf. Appuhamillage et al. [1]. Itô & McKean assert that this process is a diffusion, and provide its scale function

$$\mathfrak{p}(x) \;:=\; (1-\alpha)\, x\, \mathbf{1}_{(0,\infty)}(x) + \alpha\, x\, \mathbf{1}_{(-\infty,0]}(x)\,, \qquad x \in \mathbb{R} \tag{1}$$

and speed measure $\mathfrak{m}(\mathrm{d}x) = (2/\mathfrak{p}'(x))\,\mathrm{d}x$ for $x \neq 0$, $\mathfrak{m}(\{0\}) = 0$. The function $\mathfrak{p}(\cdot)$ is the primitive, with $\mathfrak{p}(0) = 0$, of the left-continuous step function

$$\mathfrak{p}'(x) \;:=\; (1-\alpha)\, \mathbf{1}_{(0,\infty)}(x) + \alpha\, \mathbf{1}_{(-\infty,0]}(x)\,, \qquad x \in \mathbb{R}\,;$$

and

$$\mathfrak{p}''(\mathrm{d}x) \;:=\; \big(\mathfrak{p}'(0+) - \mathfrak{p}'(0-)\big)\, \boldsymbol{\delta}_0(\mathrm{d}x) \;=\; \big(1 - 2\,\alpha\big)\, \boldsymbol{\delta}_0(\mathrm{d}x)$$

can be thought of as "second derivative measure" for this scale function. We consider also the inverse of the function $\mathfrak{p}(\cdot)$ with the property $\mathfrak{p}(\mathfrak{q}(y)) = y$, namely,

$$\mathfrak{q}(y) \;:=\; \frac{y}{1-\alpha}\, \mathbf{1}_{(0,\infty)}(y) + \frac{y}{\alpha}\, \mathbf{1}_{(-\infty,0]}(y)\,, \qquad y \in \mathbb{R}\,. \tag{2}$$

2 Construction via Changes of Time and Scale

It is very well known, of course, how to construct a diffusion with given scale function and speed measure. Following Harrison & Shepp [17] we deploy first a *time-change*, to construct a diffusion $Y(\cdot)$ in natural scale with dispersion function

$$\mathfrak{s}(y) \;:=\; \mathfrak{p}'\big(\mathfrak{q}(y)\big) \;=\; \frac{1}{\mathfrak{q}'(y)} \;=\; (1-\alpha)\, \mathbf{1}_{(0,\infty)}(y) + \alpha\, \mathbf{1}_{(-\infty,0]}(y)\,, \qquad y \in \mathbb{R} \tag{3}$$

in accordance with the celebrated method of "removal of drift" in Zvonkin [43]; then carry out the *change of scale* $X(\cdot) = \mathfrak{q}(Y(\cdot))$, to obtain the process we really want.

Indeed, the positive function $\mathfrak{s}(\cdot)$ just introduced is monotone, positive, and bounded away from the origin; as a result, the stochastic integral equation

$$Y(t) \;=\; \mathfrak{p}(x_0) + \int_0^t \mathfrak{s}\big(Y(s)\big)\, \mathrm{d}B(s)\,, \qquad 0 \le t < \infty \tag{4}$$

has a pathwise unique, strong solution, with $X(0) = x_0 \in \mathbb{R}$ an arbitrary initial condition for our skew Brownian motion and with $B(\cdot)$ a given standard one-dimensional Brownian motion (cf. Nakao [29], Le Gall [25]). In particular, we have the filtration identity

$$\mathfrak{F}^Y(t) \;=\; \mathfrak{F}^B(t)\,, \qquad 0 \le t < \infty\,. \tag{5}$$

The equation (4) describes the diffusive motion of a particle, whose local dispersion is the constant $1-\alpha$ on the half-line $(0,\infty)$, and the constant α on the half-line $(-\infty,0]$.

The solution of the equation (4) can be described "weakly" via time-change, as follows: starting a standard, one-dimensional Brownian motion $\boldsymbol{\beta}(\cdot)$ at $\boldsymbol{\beta}(0)=\mathfrak{p}(x_0)$, we introduce the continuous, strictly increasing process

$$A(\theta) := \int_0^\theta \frac{\mathrm{d}u}{\mathfrak{s}^2\big(\boldsymbol{\beta}(u)\big)}\,, \qquad 0 \le \theta < \infty$$

with $A(0)=0$, $A(\infty)=\infty$, adapted to the filtration $\big(\mathfrak{F}^{\boldsymbol{\beta}}(u)\big)_{0\le u<\infty}$ generated by $\boldsymbol{\beta}(\cdot)$; as well as the "inverse clock" $T(\cdot)$, with $A(T(t))=t$, $0\le t<\infty$. Clearly, each random variable

$$T(t)=\inf\{\theta\ge 0 : A(\theta)>t\}$$

is a stopping time of the Brownian filtration $\big(\mathfrak{F}^{\boldsymbol{\beta}}(u)\big)_{0\le u<\infty}$. We differentiate to obtain $T'(t)=\mathfrak{s}^2\big(\boldsymbol{\beta}(T(t))\big)$, then introduce the process

$$Y(t) := \boldsymbol{\beta}\big(T(t)\big)\,, \qquad 0\le t<\infty\,. \tag{6}$$

This is a square-integrable martingale of the time-changed Brownian filtration

$$\mathfrak{F}^{Y}(t) := \sigma\big(Y(s): 0\le s\le t\big) = \sigma\big(\boldsymbol{\beta}(u): 0\le u\le T(t)\big) = \mathfrak{F}^{\boldsymbol{\beta}}\big(T(t)\big)\,, \quad 0\le t<\infty\,,$$

with quadratic variation

$$\langle Y\rangle(t) \;=\; T(t) \;=\; \int_0^t \mathfrak{s}^2\big(\boldsymbol{\beta}(T(s))\big)\,\mathrm{d}s \;=\; \int_0^t \mathfrak{s}^2\big(Y(s)\big)\,\mathrm{d}s\,.$$

In particular, this square-integrable martingale $Y(\cdot)$ of (6) satisfies the *Time-Change Integral Equation* of Doeblin type (Chapter 6 in Ethier & Kurtz [10])

$$Y(t)=\boldsymbol{\beta}\left(\int_0^t \mathfrak{s}^2\big(Y(s)\big)\,\mathrm{d}s\right), \qquad 0\le t<\infty\,. \tag{7}$$

It is also "pure", in the sense of Stroock & Yor [37], [38]:

$$\bigcup_{0\le t<\infty} \mathfrak{F}^{Y}(t) \;=:\; \mathfrak{F}^{Y}(\infty) \;\equiv\; \mathfrak{F}^{\boldsymbol{\beta}}(\infty) \;:=\; \bigcup_{0\le t<\infty} \mathfrak{F}^{\boldsymbol{\beta}}(t)\,.$$

Namely, its paths can be constructed entirely from those of the Brownian Motion $\boldsymbol{\beta}(\cdot)$ in (6); and vice-versa.

In the spirit of the Doob representation of such martingales in terms of stochastic integrals, we construct then the process

$$B(t) := \int_0^t \frac{\mathrm{d}\,Y(s)}{\mathfrak{s}\big(Y(s)\big)} \;=\; \int_0^{T(t)} \frac{\mathrm{d}\,Y(A(u))}{\mathfrak{s}\big(Y(A(u))\big)} \;=\; \int_0^{T(t)} \frac{\mathrm{d}\,\boldsymbol{\beta}(u)}{\mathfrak{s}(\boldsymbol{\beta}(u))}\,, \qquad 0\le t<\infty\,, \tag{8}$$

which is standard Brownian motion on account of the P. Lévy characterization, as in Liptser & Shiryaev [27] (because it has quadratic variation

$$\langle B \rangle(t) = \int_0^{T(t)} \frac{\mathrm{d}\, u}{\mathfrak{s}^2(\boldsymbol{\beta}(u))} = A(T(t)) = t\,, \qquad 0 \le t < \infty$$

and is a local martingale of the time-changed Brownian filtration $\big(\mathfrak{F}^Y(t)\big)_{0 \le t < \infty}$ introduced above). With the processes $B(\cdot)$ as in (8), and $Y(\cdot)$ as in (6), the equation (4) is clearly satisfied. The process $Y(\cdot)$ is a diffusion in natural scale, with speed measure $\mathfrak{n}(\mathrm{d}y) = 2\,(\mathfrak{s}(y))^{-2}\,\mathrm{d}y$.

Having constructed a solution $Y(\cdot)$ of the integral equation (4), we introduce the process $X(\cdot)$ we want via

$$X(t) \;:=\; \mathfrak{q}\big(Y(t)\big)\,, \quad \text{and note the bijection} \qquad Y(t) \;=\; \mathfrak{p}\big(X(t)\big) \tag{9}$$

which is valid for each $t \in [0,\infty)$. The resulting process $X(\cdot)$ is a one-dimensional diffusion with scale function $\mathfrak{p}(\cdot)$ as in (1), and speed measure $\mathfrak{m}(\mathrm{d}x) = (2/\mathfrak{p}'(x))\,\mathrm{d}x$ for $x \neq 0$, $\mathfrak{m}(\{0\}) = 0$; its generator

$$\mathfrak{G} f(x) = \frac{1}{2}\, f''(x)\,, \;\; x \neq 0\,, \qquad \mathfrak{G} f(0) = \frac{1}{2}\, f''(0+) = \frac{1}{2}\, f''(0-)$$

acts on functions in $\mathcal{C}(\mathbb{R}) \cap \mathcal{C}^2(\mathbb{R} \setminus \{0\})$ that satisfy

$$f''(0+) \;=\; f''(0-)\,, \qquad (1-\alpha)\, f'(0-) \;=\; \alpha\, f'(0+)\,. \tag{10}$$

The scale $\mathfrak{p}(\cdot)$ of (1) is one such function.

Remark 1. From the above construction and the theory of semimartingale local time, it is fairly straightforward that both processes $X(\cdot)\,, Y(\cdot)$ satisfy the "non-stickiness" condition

$$\int_0^\infty \mathbf{1}_{\{Y(t)=0\}}\,\mathrm{d}t \;=\; \int_0^\infty \mathbf{1}_{\{X(t)=0\}}\,\mathrm{d}t \;=\; 0\,, \qquad \text{a.s.} \tag{11}$$

3 A Stochastic Equation for the Skew Brownian Motion

What are the stochastic dynamics of the process $X(\cdot) \;:=\; \mathfrak{q}(Y(\cdot))$ in (9)? Let us apply the Itô-Tanaka formula to this expression, and obtain the semimartingale decomposition

$$X(\cdot) - x_0 \;=\; \int_0^{\cdot} \mathfrak{q}'\big(Y(t)\big)\,\mathrm{d}Y(t) + \int_{\mathbb{R}} L^Y(\cdot\,, y)\,\mathfrak{q}''(\mathrm{d}y) \;=\; B(\cdot) + \frac{2\,\alpha - 1}{\alpha\,(1-\alpha)}\, L^Y(\cdot) \tag{12}$$

of $X(\cdot)$ in conjunction with (3) and the dynamics of (4).

Here and below, we denote by

$$L^{\Xi}(\cdot\,,a) \;=\; \lim_{\varepsilon\downarrow 0}\,\frac{1}{2\,\varepsilon}\int_0^{\cdot}\mathbf{1}_{\{a\le\Xi(t)<a+\varepsilon\}}\,\mathrm{d}\langle\Xi\rangle(t) \tag{13}$$

the (right-continuous) *local time* of a continuous semimartingale $\Xi(\cdot)$ at the site $a\in\mathbb{R}$, and use the shorthand

$$L^{\Xi}(\cdot) := L^{\Xi}(\cdot\,,0) \equiv L^{\Xi}(\cdot\,,0+) \tag{14}$$

for the local time at the origin. We note also, for the function of (2), the computation of its second derivative measure

$$\mathfrak{q}''(\mathrm{d}y) \;=\; \big(\mathfrak{q}'(0+)-\mathfrak{q}'(0-)\big)\,\boldsymbol{\delta}_0(\mathrm{d}y) \;=\; \frac{2\,\alpha-1}{\alpha\,(1-\alpha)}\,\boldsymbol{\delta}_0(\mathrm{d}y)\,.$$

It remains to express the local time at the origin of the semimartingale $Y(\cdot)$ in (4), in terms of the local time at the origin of the semimartingale $X(\cdot)$: noting

$$\mathrm{d}\langle Y\rangle(t) \;=\; \mathfrak{s}^2\big(Y(t)\big)\,\mathrm{d}t \;=\; \Big((1-\alpha)^2\,\mathbf{1}_{\{Y(t)>0\}}+\alpha^2\,\mathbf{1}_{\{Y(t)\le 0\}}\Big)\,\mathrm{d}t\,,$$

$\mathrm{d}\langle X\rangle(t)=\mathrm{d}t$, and $Y(t)=\mathfrak{p}(X(t))=(1-\alpha)X(t)$ on $\{Y(t)>0\}$, and recalling (11), we obtain

$$\begin{aligned} L^{Y}(\cdot\,) &= \lim_{\varepsilon\downarrow 0}\,\frac{1}{2\,\varepsilon}\int_0^{\cdot}\mathbf{1}_{\{0<Y(t)<\varepsilon\}}\,\mathrm{d}\langle Y\rangle(t) \\ &= (1-\alpha)\cdot\lim_{\varepsilon\downarrow 0}\,\frac{1-\alpha}{2\,\varepsilon}\int_0^{\cdot}\mathbf{1}_{\{0<X(t)<\frac{\varepsilon}{1-\alpha}\}}\,\mathrm{d}\langle X\rangle(t)\,. \end{aligned}$$

From this, we deduce the identity for local times at the origin

$$L^{Y}(\cdot\,) = (1-\alpha)\,L^{X}(\cdot)\,, \tag{15}$$

and from (12) the celebrated Harrison-Shepp [17] stochastic equation

$$X(t) \;=\; x_0 + B(t) + \frac{2\,\alpha-1}{\alpha}\,L^{X}(t)\,, \qquad 0\le t<\infty \tag{16}$$

for the skew Brownian motion; cf. Remark 3.

This equation is strongly solvable; indeed, $X(t)$ and $Y(t)$ are bijections of each other for each $t\in[0,\infty)$ (recall (9)), and the equation (4) is strongly solvable as noted in (5), so we have the filtration identities $\mathfrak{F}^X(t)=\mathfrak{F}^Y(t)=\mathfrak{F}^B(t)\,,\;\; 0\le t<\infty\,.$

3.1 Local Time Considerations

Three years before the work [17], John Walsh published the very important paper [40] on the Skew Brownian Motion. He computed the scale function and speed measure of this process, as well as its transition probabilities (cf. section 4 below); and studied the process not only as a diffusion, but also as a semimartingale. And at a time when local time was still not very well understood, he highlighted Skew Brownian Motion as the first nontrivial example of a continuous semimartingale, whose local time exhibits a discontinuity in the space variable — at the origin. Then HARRISON & SHEPP highlighted in [17] the fact that such discontinuities arise as a result of singularities (with respect to LEBESGUE measure) of the measure associated with the semimartingale's finite variation component.

Indeed, it is straightforward to verify, in the notation of (13), that we have

$$2\,L^{\Xi}(\cdot) - L^{|\Xi|}(\cdot) \;=\; L^{\Xi}(\cdot) - L^{\Xi}(\cdot\,,0-) \;=\; \int_0^{\cdot} \mathbf{1}_{\{\Xi(t)=0\}}\,\mathrm{d}\Xi(t)$$

for a continuous semimartingale $\Xi(\cdot)$; see, for instance, Theorem (1.7), Chapter VI in REVUZ-YOR [34], but also note that these authors use a different normalization for semimartingale local time than the one we deploy here in (13). For the skew Brownian motion $X(\cdot)$, and in conjunction with (16), the last term of the above expression is equal to $(2\,\alpha-1)\,L^X(\cdot)\,/\,\alpha$, and this gives

$$L^X(\cdot\,,0-) \;=\; \frac{1-\alpha}{\alpha}\,L^X(\cdot)\,, \qquad L^{|X|}(\cdot) \;=\; \frac{1}{\alpha}\,L^X(\cdot) \;=\; 2\,\widehat{L}^X(\cdot)\,. \tag{17}$$

We have denoted here the *"symmetric local time at the origin"* by

$$\widehat{L}^X(\cdot) \;:=\; \frac{1}{2}\Big(L^X(\,\cdot\,,0-) + L^X(\,\cdot\,,0+)\Big) \;=\; \frac{1}{2\,\alpha}\,L^X(\cdot) \;=\; \frac{1}{2}\,L^{|X|}(\cdot)\,. \tag{18}$$

It is fairly easy at this point to check the intuitively obvious: *"the absolute value of the skew Brownian motion $X(\cdot)$ is reflecting Brownian motion"*. Indeed, the ITÔ-TANAKA formula once again, this time in conjunction with (16), gives

$$|X(\cdot)| - |x_0| \;=\; \int_0^{\cdot} \mathrm{sgn}(X(t))\,\mathrm{d}X(t) + 2\,L^X(\cdot) \;=\; W(\cdot) - \frac{2\,\alpha-1}{\alpha}\,L^X(\cdot) + 2\,L^X(\cdot)$$

with the conventions

$$\mathrm{sgn}(x) := \mathbf{1}_{(0,\infty)}(x) - \mathbf{1}_{(-\infty,0]}(x)\,, \qquad W(\cdot) := \int_0^{\cdot} \mathrm{sgn}(X(t))\,\mathrm{d}B(t)\,. \tag{19}$$

From the theory of the SKOROKHOD reflection problem, we see now that the process

$$|X(\cdot)| \;=\; |x_0| + W(\cdot) + L^{|X|}(\cdot) \tag{20}$$

is Brownian motion started at $|x_0|$ and reflected at the origin.

The equation (16) and the identities

$$L^X(\cdot) \;=\; L^X(\cdot\,,0+) \;=\; \alpha\,L^{|X|}(\cdot)\,, \qquad L^X(\cdot\,,0-) \;=\; (1-\alpha)\,L^{|X|}(\cdot)\,, \tag{21}$$

consequences of (17), capture, perhaps better than anything else, the essence of this motion. It is clear from (16) that $X(\cdot)$ behaves exactly like Brownian Motion when away from the origin. Whereas standard Brownian motion $B(\cdot)$ splits the local time at the origin of its reflection $|B(\cdot)|$ with equal weights

$$L^B(\cdot\,,0+) = L^B(\cdot\,,0-) = \frac{1}{2}\,L^{|B|}(\cdot)$$

to the left and to the right of the origin, its skew counterpart $X(\cdot)$ splits the local time at the origin of the reflecting Brownian motion $|X(\cdot)|$ with weights α and $1-\alpha$ to the positive and negative sides, respectively, in the manner of (21).

We have also the additional Itô-Tanaka formulas, analogues of (20):

$$X^+(\cdot) - x_0^+ - \int_0^{\cdot} \mathbf{1}_{\{X(t)>0\}}\,\mathrm{d}B(t) \;=\; L^X(\cdot) \;=\; L^X(\cdot\,,0+) \;=\; \alpha\, L^{|X|}(\cdot)\,, \tag{22}$$

$$X^-(\cdot) - x_0^- + \int_0^{\cdot} \mathbf{1}_{\{X(t)\le 0\}}\,\mathrm{d}B(t) \;=\; \frac{1-\alpha}{\alpha}\,L^X(\cdot) \;=\; L^X(\cdot\,,0-) \;=\; (1-\alpha)\, L^{|X|}(\cdot)\,. \tag{23}$$

Together with (20), these formulas echo the original Itô–McKean idea, of generating the skew Brownian motion starting with the excursions away from the origin of a reflected Brownian motion; and changing the sign of each excursion with probability $1-\alpha$, independently from excursion to excursion (while keeping the excursion as is, namely positive, with probability α).

Remark 2. To the best of our knowledge, the Harrison-Shepp equation (16) was the first stochastic equation to be studied, that involved the local time of the unknown process. Shortly thereafter Weinryb [41], Le Gall [25]-[26] and Engelbert-Schmidt [13] began the study of more general such equations of the form

$$X(\cdot) \;=\; x_0 + \int_0^{\cdot} \sigma(X(t))\,\mathrm{d}B(t) + \int_{\mathbb{R}} L^X(\cdot\,,\xi)\,\boldsymbol{\nu}(\mathrm{d}\xi)\,, \tag{24}$$

for appropriate measurable functions $\sigma(\cdot)$ and measures $\boldsymbol{\nu}$ on $\mathcal{B}(\mathbb{R})$; we refer to the survey papers by Lejay [24] and Ouknine [31].

Remark 3. In terms of the symmetric version of local time at the origin introduced in (18), the equation (16) takes the form

$$X(t) \;=\; x_0 + B(t) + 2\,\big(2\,\alpha - 1\big)\,\widehat{L}^X(t)\,, \qquad 0 \le t < \infty \tag{25}$$

as it appears in Harrison & Shepp [17], as well as in Appuhamillage et al. [1], Borodin & Salminen [8].

Again with the symmetric local time (18), Barlow et al. [4] study the generalization

$$X(t) \;=\; x_0 + B(t) + \psi\big(\,\widehat{L}^X(t)\big)\,, \qquad 0 \le t < \infty \tag{26}$$

of the stochastic equation (25), and show that it too admits a pathwise unique, strong solution, for every function $\psi : \mathbb{R} \to \mathbb{R}$ of class C^1 with $\psi(0) = 0$, $-2 < \psi'(\cdot) < 2$. These authors call the resulting process $X(\cdot)$ "variably-skewed Brownian Motion", and note its connections to Dynamic Allocation ("multi-armed bandit") problems. A time-inhomogeneous version of (26) is studied by ÉTTORÉ - MARTINEZ [11].

Remark 4. It follows from (7), (9) that the Skew Brownian Motion $X(\cdot)$ satisfies, in addition to the HARRISON-SHEPP equation (16), also the *Time Change Integral Equation*

$$X(t) = \big(\mathfrak{q} \circ \boldsymbol{\beta}\big) \left(\int_0^t \mathfrak{s}^2 \big(X(s)\big)\, \mathrm{d}s \right), \qquad 0 \le t < \infty \tag{27}$$

of DOEBLIN-type, driven by the Brownian Motion $\beta(\cdot)$ started at $\boldsymbol{\beta}(0) = \mathfrak{p}(x_0)$. Here $\mathfrak{q}(\cdot)$ is the the inverse scale function of (2), and $\mathfrak{s}(\cdot)$ the dispersion function in (3). We note that the above equation is completely devoid of local times.

3.2 The Same Steps, now in Reverse

We have shown how to derive the equation (16) for the skew Brownian motion $X(\cdot)$, from the "driftless" stochastic differential equation of its "naturally scaled" version

$$Y(\cdot) \;=\; \mathfrak{p}(X(\cdot)) \;=\; \big(1-\alpha\big) X^+(\cdot) - \alpha\, X^-(\cdot)\,. \tag{28}$$

This works also in reverse: starting with the HARRISON-SHEPP equation (16) and deploying the ITÔ - TANAKA formula yet again, we obtain

$$Y(\cdot) - \mathfrak{p}(x_0) \;=\; \int_0^\cdot \mathfrak{p}'\big(X(t)\big) \left[\mathrm{d}B(t) + \frac{2\,\alpha - 1}{\alpha}\, \mathrm{d}L^X(t) \right] + \big(\mathfrak{p}'(0+) - \mathfrak{p}'(0-)\big)\, L^X(\cdot)\,.$$

The local times vanish from this equation because of the identity (*"jump condition"*)

$$(1-\alpha)\, \mathfrak{p}'(0-) \;=\; \alpha\, \mathfrak{p}'(0+)\,,$$

and we arrive at (4) once we recall $\mathfrak{s}(\cdot) = \mathfrak{p}'(\mathfrak{q}(\cdot))$ from (3).

Indeed, if we insist that the scale function $\mathfrak{p}(\cdot)$ be linear on both $(-\infty, 0)$ and $(0, \infty)$ (corresponding to the notion that the process behaves like Brownian motion away from the origin), and pin it at $\mathfrak{p}(0) = 0$, then the choice in (1) is forced on us by the above jump condition.

• A bit more generally, let us take any function $f(\cdot)$ in $\mathcal{C}(\mathbb{R}) \cap \mathcal{C}^2(\mathbb{R} \setminus \{0\})$ with the properties (10). Applying the generalized ITÔ rule in conjunction with the condition (11) and the equation (16), we deduce the local martingale property of the process

$$f\big(X(\cdot)\big) - f(x_0) - \int_0^\cdot \mathfrak{G} f\big(X(t)\big)\, \mathrm{d}t \;=\; f\big(X(\cdot)\big) - f(x_0) - \frac{1}{2} \int_0^\cdot f''\big(X(t)\big)\, \mathrm{d}t$$

$$= \int_0^\cdot D^- f\big(X(t)\big)\, \mathrm{d}X(t) + \big(\, f'(0+) - f'(0-)\,\big)\, L^X(\cdot)$$

$$= \int_0^{\cdot} D^- f\big(X(t)\big)\, \mathrm{d}B(t) + \left(\frac{2\,\alpha - 1}{\alpha}\, f'(0-) + f'(0+) - f'(0-) \right) L^X(\cdot)$$

$$= \int_0^{\cdot} D^- f\big(X(t)\big)\, \mathrm{d}B(t)\,. \tag{29}$$

Now suppose that, in addition, we want $\mathfrak{G} f(\cdot) \equiv 0$ on $\mathbb{R} \setminus \{0\}$, so that $f\big(X(\cdot)\big)$ is a local martingale itself. This forces $f''(\cdot) \equiv 0$ away from the origin, so $f(x) = \beta_\pm\, x + \gamma_\pm$ for $x \in \mathbb{R}_\pm$; and if we insist that $f(\cdot)$ be not just continuous at the origin, but actually nail its value $f(0) = 0$ there, we get $\gamma_\pm = 0$. On the other hand, the jump condition

$$(1-\alpha)\, f'(0-) = \alpha\, f'(0+)$$

gives $(1-\alpha)\,\beta_- = \alpha\,\beta_+$, which we can manage by taking $\beta_- = \alpha$, $\beta_+ = 1 - \alpha$, or equivalently

$$f(x) \;=\; (1-\alpha)\, x\, \mathbf{1}_{(0,\infty)}(x) + \alpha\, x\, \mathbf{1}_{(-\infty,0]}(x) \;=\; \mathfrak{p}(x)\,, \qquad x \in \mathbb{R}\,;$$

that is, the function of (1). Back into (29), this leads to the equation of (4), namely

$$Y(\cdot) \,-\, y_0 \;=\; \mathfrak{p}\big(X(\cdot)\big) - \mathfrak{p}(x_0) = \int_0^{\cdot} \mathfrak{p}'\big(X(t)\big)\, \mathrm{d}B(t)$$

$$= \int_0^{\cdot} \mathfrak{p}'\big(\mathfrak{q}(Y(t))\big)\, \mathrm{d}B(t) = \int_0^{\cdot} \mathfrak{s}\big(Y(t)\big)\, \mathrm{d}B(t)\,.$$

4 Transition Probabilities

Let us denote by

$$\boldsymbol{\varphi}_t(\xi) \;=\; \frac{1}{\sqrt{2\,\pi\, t}}\, \exp\left\{ -\frac{\xi^2}{2\,t} \right\}, \qquad t > 0\,, \quad \xi \in \mathbb{R}^2$$

the standard Gaussian kernel. In terms of it, the transition probabilities of skew Brownian motion

$$\mathbb{P}\big(X(t) \in \mathrm{d}y \,|\, X(0) = x\big) \;=\; \mathfrak{p}_t^{(\alpha)}(x,y)\, \mathrm{d}y$$

with skewness parameter $\alpha \in (0,1)$ as in (16), are given as

$$\mathfrak{p}_t^{(\alpha)}(0,y) \;=\; \Big(2\,\alpha \cdot \mathbf{1}_{(0,\infty)}(y) + 2\,(1-\alpha) \cdot \mathbf{1}_{(-\infty,0)}(y) \Big)\, \boldsymbol{\varphi}_t(y)\,, \qquad y \in \mathbb{R}$$

when starting at the origin, and more generally (see Walsh [40] for the derivations) as

$$\mathfrak{p}_t^{(\alpha)}(x,y) \;=\; \boldsymbol{\varphi}_t(y-x) + (2\,\alpha - 1)\, \boldsymbol{\varphi}_t(y+x)\,, \quad \text{for} \quad x > 0,\; y > 0$$

$$\mathfrak{p}_t^{(\alpha)}(x,y) \,=\, \varphi_t(y-x) + (1-2\,\alpha)\,\boldsymbol{\varphi}_t(y+x)\,, \quad \text{for} \quad x<0,\ y<0$$
$$\mathfrak{p}_t^{(\alpha)}(x,y) \,=\, 2\,\alpha\,\boldsymbol{\varphi}_t(y-x)\,, \quad \text{for} \quad x<0,\ y>0$$
$$\mathfrak{p}_t^{(\alpha)}(x,y) \,=\, 2\,(1-\alpha)\,\boldsymbol{\varphi}_t(y-x)\,, \quad \text{for} \quad x>0,\ y<0\,.$$

Appuhamillage et al. [1] compute the joint distribution of Skew Brownian Motion $X(t)$ and its symmetric local time at the origin $\widehat{L}^X(t)$, for arbitrary $t \in (0,\infty)$.

5 An Extended Tanaka Equation

In another important development, Barlow [2] uses the description (16) of Skew Brownian Motion, to study the extension

$$Z(\cdot) \,=\, z + \int_0^{\cdot} \boldsymbol{\varrho}\big(Z(t)\big)\,\mathrm{d}W(t)\,, \qquad \boldsymbol{\varrho}(\xi) \,:=\, \beta\,\mathbf{1}_{(0,\infty)}(\xi) - \gamma\,\mathbf{1}_{(-\infty,0]}(\xi) \tag{30}$$

of the classical Tanaka equation (cf. Liptser & Shiryaev [27]) with arbitrary constants $\beta > 0$, $\gamma > 0$; in particular, he shows that pathwise uniqueness – thus also strong solvability – fails for this equation.

The idea of Barlow's proof is to start with the Skew Brownian Motion $X(\cdot)$ of (16), introduce by analogy with (28) the process

$$Z(\cdot) \,:=\, \beta\,X^+(\cdot) - \gamma\,X^-(\cdot)\,, \tag{31}$$

and invoke the Itô-Tanaka formulae (22), (23) to obtain

$$\begin{aligned} Z(\cdot) - Z(0) \,&=\, \int_0^{\cdot} \big(\beta\,\mathbf{1}_{\{X(t)>0\}} + \gamma\,\mathbf{1}_{\{X(t)\le 0\}}\big)\,\mathrm{d}X(t) + (\beta-\gamma)\,L^X(\cdot) \\ &= \int_0^{\cdot} \boldsymbol{\varrho}\big(X(t)\big)\,\operatorname{sgn}\big(X(t)\big)\left[\mathrm{d}B(t) + \frac{2\,\alpha-1}{\alpha}\,\mathrm{d}L^X(t)\right] + (\beta-\gamma)\,L^X(\cdot) \\ &= \int_0^{\cdot} \boldsymbol{\varrho}\big(Z(t)\big)\,\mathrm{d}W(t) + \left(\frac{2\,\alpha-1}{\alpha}\,\gamma + \beta - \gamma\right) L^X(\cdot)\,. \end{aligned} \tag{32}$$

We have recalled here the notation of (19), and noted

$$\operatorname{sgn}\big(X(t)\big) \,=\, \operatorname{sgn}\big(Z(t)\big)\,, \qquad \boldsymbol{\varrho}\big(X(t)\big) \,=\, \boldsymbol{\varrho}\big(Z(t)\big)\,.$$

If we select

$$\alpha \,=\, \frac{\gamma}{\beta+\gamma}$$

as the skewness parameter, the local time terms disappear from the display (32); this takes then the form of (30) with initial condition $z = \beta\,x_0^+ - \gamma\,x_0^-$ and, by analogy with (19), driving Brownian motion $W(\cdot) \,=\, \int_0^{\cdot} \operatorname{sgn}(X(t))\,\mathrm{d}B(t) \,=\, |X(\cdot)| - |x_0| - L^{|X|}(\cdot)$.

Consider now two distinct skew Brownian motions $X_1(\cdot)$, $X_2(\cdot)$ with $|X_1(\cdot)| \equiv |X_2(\cdot)| \,=:\, R(\cdot)$; we obtain them both by flipping the excursions of the reflecting

Brownian motion $R(\cdot)$ independently from one another, upwards with probability $\alpha = \gamma/(\beta+\gamma)$ and downwards with probability $1-\alpha = \beta/(\beta+\gamma)$. These two distinct skew Brownian motions $X_1(\cdot)$, $X_2(\cdot)$ generate, via the recipe (31), distinct solutions $Z_1(\cdot)$, $Z_2(\cdot)$ of the equation (30) driven by the *same* Brownian motion $W(\cdot) = R(\cdot) - |x_0| - L^R(\cdot)$. It follows that *pathwise uniqueness fails for the equation* (30), just as it fails for its counterpart with $\beta = \gamma = 1$, the original TANAKA equation.

By contrast, it has been well-known since the work of NAKAO [29] (see also LE GALL [25]), that the stochastic equation

$$Z(\cdot) \;=\; z + \int_0^{\cdot} \boldsymbol{\sigma}\big(Z(t)\big)\,\mathrm{d}W(t)\,, \qquad \boldsymbol{\sigma}(\xi) \;:=\; \beta\,\mathbf{1}_{(0,\infty)}(\xi) + \gamma\,\mathbf{1}_{(-\infty,0]}(\xi)$$

with arbitrary real constants $\beta > 0$, $\gamma > 0$, has a pathwise unique, strong solution.

Remark 5. Theorem 8.1 in FERNHOLZ ET AL. [15] shows that the addition of a perturbation to (30), in the form of a Brownian motion $V(\cdot)$ independent of $W(\cdot)$, restores pathwise uniqueness to the resulting *extended* TANAKA *equation*

$$Z(\cdot) \;=\; z + \int_0^{\cdot} \boldsymbol{\varrho}\big(Z(t)\big)\,\mathrm{d}W(t) + V(\cdot)\,;$$

see also PROKAJ [33]. This answers a question raised by Marc YOR in 2010.

6 A Planar Version of (4): Dispersions by Rank

The paper [15] considers, among several other things, the planar analogue

$$Y_1(\cdot) = y_1 + \int_0^{\cdot} \Big(\varrho\,\mathbf{1}_{\{Y_1(t)>Y_2(t)\}} + \sigma\,\mathbf{1}_{\{Y_1(t)\le Y_2(t)\}} \Big)\,\mathrm{d}B_1(t) \tag{33}$$

$$Y_2(\cdot) = y_2 + \int_0^{\cdot} \Big(\varrho\,\mathbf{1}_{\{Y_1(t)\le Y_2(t)\}} + \sigma\,\mathbf{1}_{\{Y_1(t)>Y_2(t)\}} \Big)\,\mathrm{d}B_2(t) \tag{34}$$

of the stochastic integral equation (4). Here $\varrho\,,\sigma$ are positive constants satisfying $\varrho^2 + \sigma^2 = 1$, $(y_1, y_2) \in \mathbb{R}^2$, and $B_1(\cdot)$, $B_2(\cdot)$ are independent standard Brownian Motions. This system describes the motions of two purely diffusive particles, driven by independent Brownian motions, *whose local dispersions are assigned by rank:* $\varrho > 0$ for the "leader", $\sigma > 0$ for the "laggard", with $\varrho^2 + \sigma^2 = 1$.

It is shown in [15] that the above system (33), (34) *admits a strong solution, which is pathwise unique.* Furthermore, it is straightforward to check that the difference $Y(\cdot) := Y_1(\cdot) - Y_2(\cdot)$ is standard Brownian Motion; in particular, the set of time-points $\{t \ge 0 : Y_1(t,\omega) = Y_2(t,\omega)\}$ at which the two particles are "tied", has zero LEBESGUE measure for $\mathbb{P}-$a.e. $\omega \in \Omega$. It is also clear that the pair of processes $\big(Y_1(\cdot), Y_2(\cdot)\big)$ are *orthogonal* square-integrable martingales (i.e., $\langle Y_1, Y_2 \rangle(\cdot) \equiv 0$) of the filtration they jointly generate, with respective quadratic variations

$$\langle Y_1 \rangle(\cdot) = \int_0^{\cdot} \Big(\varrho^2 \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} + \sigma^2 \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} \Big) \, \mathrm{d}t =: T_1(\cdot) \, ,$$

$$\langle Y_2 \rangle(\cdot) = \int_0^{\cdot} \Big(\varrho^2 \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} + \sigma^2 \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} \Big) \, \mathrm{d}t =: T_2(\cdot) \, .$$

The F. B. Knight Theorem shows at this point, that the processes

$$\boldsymbol{\beta}_1(u) := Y_1\big(A_1(u)\big) \, , \quad \boldsymbol{\beta}_2(u) := Y_2\big(A_2(u)\big) \, , \qquad 0 \le u < \infty \tag{35}$$

are *independent* standard Brownian Motions with $\boldsymbol{\beta}_1(0) = y_1$, $\boldsymbol{\beta}_2(0) = y_2$, and with $A_1(\cdot)$, $A_2(\cdot)$ the inverses of the strictly increasing functions $T_1(\cdot)$, $T_2(\cdot)$.

Whereas, inverting the time-changes in (35), the system (33)-(34) can be cast in the form of a Doeblin-type system of *Time-Change Integral Equations*

$$Y_1(\cdot) \; = \; \boldsymbol{\beta}_1 \left(\int_0^{\cdot} \Big(\varrho^2 \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} + \sigma^2 \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} \Big) \, \mathrm{d}t \right) ,$$

$$Y_2(\cdot) \; = \; \boldsymbol{\beta}_2 \left(\int_0^{\cdot} \Big(\varrho^2 \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} + \sigma^2 \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} \Big) \, \mathrm{d}t \right) ,$$

driven by the independent Brownian Motions $\boldsymbol{\beta}_1(\cdot)$, $\boldsymbol{\beta}_2(\cdot)$.

6.1 The Same System, Now With Skew-Elastic Collisions

Let us recall the notation of (14), (13) for the local time of a continuous semimartingale at the origin, and consider with Fernholz et al. [14] the so-called "skew-elastic" variant of the system (33)-(34), with appropriate pairs of real numbers (η_i, ζ_i), $i = 1, 2$:

$$\begin{aligned} Y_1(\cdot) = y_1 + \int_0^{\cdot} \Big(\varrho \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} + \sigma \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} \Big) \, \mathrm{d}B_1(t) \\ + \frac{1 - \zeta_1}{2} \, L^{Y_1 - Y_2}(\cdot) + \frac{1 - \eta_1}{2} \, L^{Y_2 - Y_1}(\cdot) \, , \end{aligned} \tag{36}$$

$$\begin{aligned} Y_2(\cdot) = y_2 + \int_0^{\cdot} \Big(\varrho \, \mathbf{1}_{\{Y_1(t) \le Y_2(t)\}} + \sigma \, \mathbf{1}_{\{Y_1(t) > Y_2(t)\}} \Big) \, \mathrm{d}B_2(t) \\ + \frac{1 - \zeta_2}{2} \, L^{Y_1 - Y_2}(\cdot) + \frac{1 - \eta_2}{2} \, L^{Y_2 - Y_1}(\cdot) \, . \end{aligned} \tag{37}$$

Under the dispensation of (36)-(37), when these particles collide, they are "dragged" by amounts proportional to the local times accumulated at the origin by the differences $Y_1 - Y_2$ and $Y_2 - Y_1$ "during the time-interval of the collision".

In terms of the parameters

$$\zeta := 1 + \frac{\zeta_1 - \zeta_2}{2}\,, \quad \overline{\zeta} := \frac{\zeta_1 + \zeta_2}{2}\,, \qquad \eta := 1 - \frac{\eta_1 - \eta_2}{2}\,, \quad \overline{\eta} := \frac{\eta_1 + \eta_2}{2}\,,$$

it was shown in [14] that the condition

$$\zeta + \eta \neq 0\,, \qquad 0 \le \alpha := \frac{\eta}{\eta + \zeta} \le 1$$

is necessary and sufficient for the above system of stochastic equations to admit a pathwise unique, strong solution.

When $\zeta_1 = \eta_1 = 1$ (resp., $\zeta_2 = \eta_2 = 1$), particle 1 crosses the trajectory of particle 2 (resp., the other way round) without "feeling" it, i.e., without being subjected to any local time drag. And when $\zeta_i = \eta_i = 1,\ i = 1, 2$ all collisions are frictionless, and we are in the realm of (33), (34).

When $\zeta = 0 \neq \eta$ (resp., $\eta = 0 \neq \zeta$), the trajectory of particle 1 bounces off that of particle 2 (resp., the other way round) as if the latter trajectory were a perfectly reflecting boundary.

When $\eta\,\overline{\zeta} + \zeta\,\overline{\eta} = 0$, the trajectory $R_2(\cdot) := Y_1(\cdot) \wedge Y_2(\cdot)$ of the "laggard" particle evolves just like Brownian Motion with local dispersion σ, and "feels no pressure" (local time drag) from the "leader" $R_1(\cdot) := Y_1(\cdot) \vee Y_2(\cdot)$, which evolves like an *independent* Brownian Motion with local dispersion ϱ and is reflected off the trajectory of the laggard. This gives a novel realization of the stochastic motions studied by Burdzy & Nualart [5], Soucaliuc et al. [35]-[36].

For other values of the parameters, collisions are neither totally frictionless nor perfectly reflecting, but asymmetric and "elastic": they are governed by local time drag in an asymmetric fashion, due to the presence of both the right- and left- local times at the origin, respectively $L^Y(\cdot) = L^Y(\,\cdot\,; 0+)$ and $L^Y(\,\cdot\,; 0-)$, for the difference $Y := Y_1 - Y_2$. We call such collisions "skew-elastic". More general such systems of interacting particles, and their relationship to TASEP and to determinantal structures, are discussed in Karatzas, Pal & Shkolnikov [21].

7 "Spider" Brownian Motion

In the epilogue of his paper on the Skew Brownian Motion, Walsh [40] speculated about the motion $Z(\cdot) = \big(R(\cdot), \Theta(\cdot)\big)$ of a similar Brownian particle, now moving on the plane (or in higher-dimensional Euclidean space, if one wishes) rather than the real line, but in *radial* fashion: only along rays emanating from the origin. (Imagine the pokes of a bicycle wheel, a "hub-and-spoke" routing system, or the legs of a spider.) He posited that, when started at a point $Z(0) = z \in \mathbb{R}^2$ away from the origin 0, the particle should move like a one-dimensional Brownian Motion along the ray joining z and 0, until the origin is reached; then be kicked away from the origin according to an entrance law that makes $R(\cdot) := |Z(\cdot)|$ reflecting Brownian Motion, while randomizing the angular part $\Theta(\cdot)$ according to a "spinning" probability measure $\boldsymbol{\nu}$ on $[0, 2\pi)$.

To quote verbatim from [40]: “The idea is to take each excursion” (of reflecting Brownian Motion) “and, instead of giving it a random sign, assign it a random variable Θ with a given distribution on $[0, 2\pi)$, and to do this independently for each excursion”. The resulting process “is a diffusion which, when away from the origin, is a Brownian Motion along a ray, but which has what one might be called a *roundhouse singularity* at the origin: when the process enters it, it, like Stephen Leacock’s hero, immediately rides off in all directions at once”.

Such a process was constructed eleven years later by Barlow, Pitman & Yor [6] via the right prescription of its transition semigroup, and was shown to have the strong Markov and Feller (though not the strong Feller) properties. We call it here Walsh *(or “spider”) Brownian Motion.* These authors also proved that the so-constructed $Z(\cdot)$ has the *martingale representation property:* every (local) martingale of the filtration this process generates, is a stochastic integral with respect to the Brownian Motion

$$W(\cdot) := R(\cdot) - R(0) - L^R(\cdot)\,,$$

constructed in the manner of (20) and adapted to the filtration generated by $Z(\cdot)$.

Now, when the spinning measure charges only $N = 2$ angles, say $\vartheta = 0$ and $\vartheta = \pi$ for concreteness, we are back to the Skew Brownian Motion whose “natural” filtration, as we have seen, not only has the martingale representation property, but actually *coincides with the filtration generated by some scalar Brownian Motion* (cf. equation (16) and the discussion following it).

Might this be the case more generally? This kind of question was first posed by Marc Yor in 1979, and was rephrased by Barlow et al. [6] in the specific context of the Walsh Brownian Motion. It received a most emphatically negative answer when, in a veritable tour-de-force, Tsirel’son [39] showed that, already for spinning measures $\boldsymbol{\nu}$ charging $N = 3$ (or more) angles, the filtration of the resulting Walsh Brownian Motion $Z(\cdot)$ cannot be generated by *any* Brownian Motion in *any* dimension. For a complete and concise exposition of this remarkable result, the reader may wish to consult the first chapter in the doctoral dissertation by Yan [42].

7.1 Semimartingale Dynamics of Harrison-Shepp Type

It is natural to ask at this point, *whether the* Walsh *Brownian Motion admits a description of the* Harrison-Shepp *type* (16).

Expressing the planar motion $Z(\cdot) = \big(R(\cdot), \Theta(\cdot)\big)$ in its Cartesian coördinates $Z_1(\cdot)$ and $Z_2(\cdot)$, Ichiba et al. [19] established the semimartingale dynamics

$$Z_i(\cdot) = z_1 + \int_0^{\cdot} \mathbf{1}_{\{Z(t)\neq 0\}}\, \mathfrak{f}_i\big(Z(t)\big)\, \mathrm{d}W(t) + \gamma_i\, L^{|Z|}(\cdot)\,, \qquad i = 1, 2 \tag{38}$$

and the “thinning of local time” properties

$$L^{Z_i}(\cdot) = \alpha_i^{(+)}\, L^{|Z|}(\cdot)\,, \qquad i = 1, 2\,, \tag{39}$$

with the notation $\gamma_i := \alpha_i^{(+)} - \alpha_i^{(-)}$ and

$$\alpha_1^{(\pm)} := \int_0^{2\pi} \big(\cos(\vartheta)\big)^{\pm} \boldsymbol{\nu}(\mathrm{d}\vartheta)\,, \qquad \mathfrak{f}_1(z) \equiv f_1(r,\vartheta) := r\cos(\vartheta)\,,$$

$$\alpha_2^{(\pm)} := \int_0^{2\pi} \big(\sin(\vartheta)\big)^{\pm} \boldsymbol{\nu}(\mathrm{d}\vartheta)\,, \qquad f_2(z) \equiv \mathfrak{f}_1(r,\vartheta) := r\sin(\vartheta)\,,$$

where ϑ denotes the angle relative to the x half-axis in Cartesian coördinates. Whereas, when $\alpha_i^{(\pm)} > 0$, we obtain from the dynamics of (38) and the thinning of (39) the "true analogue" of the Harrison-Shepp equation (16) in the present context, namely,

$$Z_i(\cdot) = z_i + \int_0^{\cdot} \mathbf{1}_{\{Z(t)\neq 0\}}\, \mathfrak{f}_i\big(Z(t)\big)\,\mathrm{d}W(t) + \left(1 - \frac{\alpha_i^{(-)}}{\alpha_i^{(+)}}\right) L^{Z_i}(\cdot)\,, \qquad i = 1,2\,. \quad (40)$$

With some extra work, a full-fledged stochastic calculus for Brownian Motions – and more generally, semimartingales – of this "Walsh/spider" type, replete with the appropriate Freidlin-Sheu [16] change-of-variable formula, is developed in [19]. This, in turn, allows the formulation, and opens the door to the resolution, of interesting stochastic optimization problems in this context; see, for instance, Karatzas & Yan [23], Bayraktar & Zhang [7], Martinez & Ohavi [28], Ohavi [30].

8 Brownian Motions with "Sticky" Sites

Another close relative of the Skew Brownian Motion is the so-called *Sticky Reflecting Brownian Motion.* This process describes the motion of a Brownian particle reflected at the origin, but also required to spend there a positive amount of time (with respect to Lebesgue measure). The simplest way to describe the process is via the stochastic integral equation

$$X(t) = x + \mu \int_0^t \mathbf{1}_{\{X(s)=0\}}\,\mathrm{d}s + \int_0^t \mathbf{1}_{\{X(s)>0\}}\,\mathrm{d}B(s)\,, \qquad 0 \le t < \infty \qquad (41)$$

driven by standard Brownian Motion $B(\cdot)$, with given initial condition $x \ge 0$ and "stickiness parameter" $\mu > 0$. Away from the origin, this is the motion of a diffusing Brownian particle. When at the origin, the motion ceases to be diffusive, and receives positive drift μ. As a result, it never becomes negative; and its occupation time at the origin is proportional to its semimartingale (right) local time there, namely,

$$\int_0^T \mathbf{1}_{\{X(t)=0\}}\,\mathrm{d}t \;=\; \frac{1}{\mu}\,L^X(T)\,.$$

A very detailed study of this and of a related process is carried out by Engelbert & Peskir [12], along with historical and bibliographical references. These authors show that the equation (41) admits a unique-in-distribution weak solution; but no strong solution, thus vindicating a conjecture of A.V. Skorokhod. In this vein, we refer also to Bass [3] and to the early work of Chitashvili [9].

Consider now the stochastic integral equation

$$X(t) = x + \lambda\, t + \int_0^t \mathbf{1}_{\{X(s)>0\}}\, \mathrm{d}B(s)\,, \qquad 0 \le t < \infty \tag{42}$$

with real constants x (initial condition) and λ (drift), driven again by Brownian Motion $B(\cdot)$. This equation is somewhat similar to (41), and constitutes a "one-sided" version of the Tanaka-like equation (30) with an additional, constant drift term. It was studied by Karatzas, Shiryaev & Shkolnikov [22], who showed that it admits:
(i) a strong solution which is unique pathwise, thus also in distribution, if $\lambda \le 0$;
(ii) a weak solution which is unique in distribution but fails to be strong, if $\lambda > 0$. This solution is also shown to be "sticky", in that the so-called "non-stickiness" condition

$$\int_0^T \mathbf{1}_{\{X(t)=0\}}\, \mathrm{d}t \;=\; 0\,, \qquad \mathbb{P}-\text{a.e.}$$

fails, for every $T \in (0, \infty)$.

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